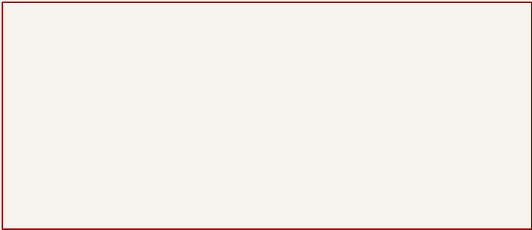


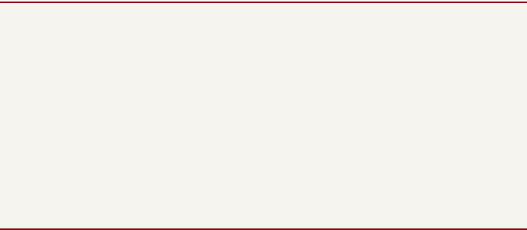
Vanchor Helm Board – sheet index

- 1 carrier: 12V in, protection, 5V buck header, Orange Pi Zero 3 ribbon header, UART/I2C JSTs
- 2 control: Pico 2, external thrust–driver interface (J13), BTN8982 servo bridge, AS5600

carrier



control



12V battery only; 5V from generic buck module on J14
Orange Pi Zero 3 (IDC ribbon) + Pico 2; external IBT–2 thrust driver; on–board servo bridge

vanchor–ng

Sheet: /
File: vanchor–helm.kicad_sch

Title: Vanchor Helm Board

Size: A3Date: 2026–07–02

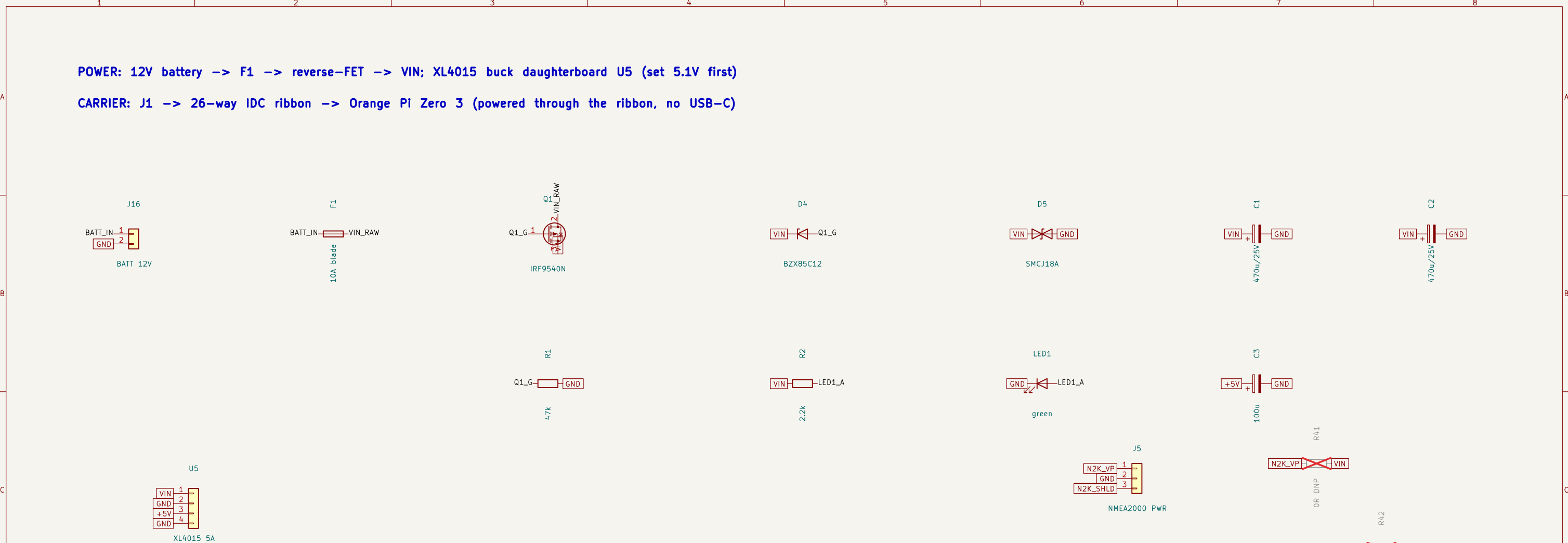
Rev: v4.1

KiCad E.D.A. 10.0.3

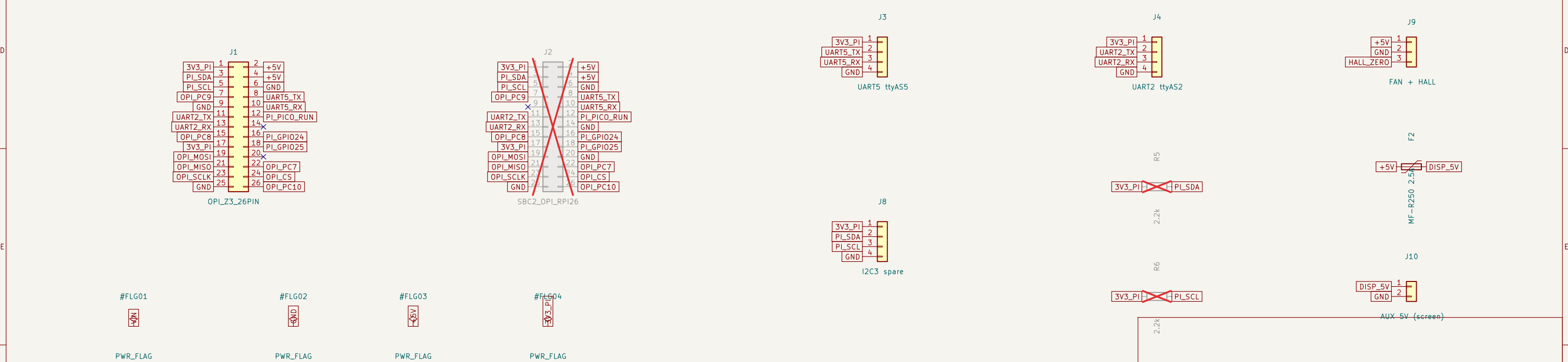
Id: 1/3

1	2	3	4	5	6	7	8
POWER: 12V battery -> F1 -> reverse-FET -> VIN; XL4015 buck daughterboard U5 (set 5.1V first)							
CARRIER: J1 -> 26-way IDC ribbon -> Orange Pi Zero 3 (powered through the ribbon, no USB-C)							

1	2	3	4	5	6	7	8
POWER: 12V battery -> F1 -> reverse-FET -> VIN; XL4015 buck daughterboard U5 (set 5.1V first)							
CARRIER: J1 -> 26-way IDC ribbon -> Orange Pi Zero 3 (powered through the ribbon, no USB-C)							



ORANGE PI ZERO 3 HEADER + UART/I2C BREAKOUTS



SERVO: 2x BTN8982TA on protected VIN; AS5600 encoder cable on J11

Sheet: /control/
File: control.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 3/3